Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. Consider the following matrix: $A = \begin{bmatrix} 11 & \frac{1}{2} & -2 & 3 \\ 4 & 1 & 1 & 0 \end{bmatrix}$.

   (a) What type of matrix is A? Is a 2 by 4 matrix (2×4), so it's a rectangular matrix

   (b) What are the entries of A at positions $(1, 2)$ and $(2, 1)$?

   - $a_{1,2} = \frac{1}{2}$
   - $a_{2,1} = 4$

   (c) At which positions in A is the entry equal to 1? In $a_{2,2}$ and $a_{2,3}$

2. Write the matrix $\left(A = [a_{ij}]_{3\times 3}\right)$ defined by:

   (a) $a_{ij} = i + j$

   $$A = \begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}$$

   (b) $a_{ij} = \begin{cases} 0, & \text{if } i = j, \\ -1, & \text{if } i > j, \\ 2, & \text{if } i < j. \end{cases}$

   $$A = \begin{pmatrix} 0 & 2 & 2 \\ -1 & 0 & 2 \\ -1 & -1 & 0 \end{pmatrix}$$

3. For each of the following parts, determine, if possible, the real numbers x and y such that the matrices A and B are equal.

   Remember: Two matrices are equal if they have the same dimensions and all their corresponding entries **are equal**.

   (a) $A = \begin{bmatrix} 1 & 0 \\ x & y \\ -1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ \frac{y}{2} & x+1 \\ -1 & 3 \end{bmatrix}$

   Solution:

- From the first row: $a_{11} = 1 = b_{11}$ and $a_{12} = 0 = b_{12}$.
- From the second row: $a_{21} = x = \frac{y}{2} = b_{21}$ and $a_{22} = y = x + 1 = b_{22}$.
- From the third row: $a_{31} = -1 = b_{31}$ and $a_{32} = 3 = b_{32}$.

The point is that rows 1 and 3 do not give us any information about x and y, so we only need to solve the system of equations from the second row:

$$\begin{cases} x = \dfrac{y}{2}, \\ y = x + 1. \end{cases}$$

Substitute $x = \frac{y}{2}$ into the second equation:

$$\begin{aligned} y &= \frac{y}{2} + 1, \\ 2y &= y + 2, \\ y &= 2, \qquad x = \frac{y}{2} = 1. \end{aligned}$$

$$x = 1, \qquad y = 2$$

(b) $A = \begin{bmatrix} x^3 & 2 & 0 \\ 3 & y & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 3x+1 & 2 & y \\ 3 & -2 & -1 \end{bmatrix}$

Solution:

(c) $A = \begin{bmatrix} 2 & -5 \\ 0 & x+y \\ 2y & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -5 \\ 0 & 2x+4 \\ y+1 & 1 \end{bmatrix}$

Solution:

——— Exercise 4 ———

4. Consider the following matrices:

$$A = \begin{bmatrix} 2 & 2 \\ 3 & 3 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 3 \\ -4 & 0 \\ 0 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & -2 \end{bmatrix}, \quad E = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & 1 \\ -2 & 2 & 4 \end{bmatrix}, \quad F = \begin{bmatrix} -2 & 1 & 3 \\ -1 & 1 & -2 \\ 0 & 1 & 3 \end{bmatrix}$$

Determine the following matrices, only if they are defined. If they are not defined, explain why.

A) $A + B$	D) $D + E$	G) $A + D$	J) AB
B) $B + A$	E) $-A + 2B$	H) $A - A$	K) AC
C) $A - B$	F) $A + C$	I) $C - C$	L) CA

M) AD P) FE S) $A(CD)$ V) $A(CD+D)$

N) DA Q) $A(3D)$ T) $(2A)(C(3D))$

O) EF R) $(AC)D$ U) $(2A)((2C)(2D))$ W) C^3

A) $\begin{bmatrix} 1 & 5 \\ -1 & 3 \\ -1 & -2 \end{bmatrix}$, B) $\begin{bmatrix} 1 & 5 \\ -1 & 3 \\ -1 & -2 \end{bmatrix}$ C) $\begin{bmatrix} 3 & -1 \\ 7 & 3 \\ -1 & 0 \end{bmatrix}$

D) Not defined, because D is a 2×3 matrix and E is a 3×3 matrix, so they do not have the same dimensions.

E) $-1 \cdot \begin{bmatrix} 2 & 2 \\ 3 & 3 \\ -1 & -1 \end{bmatrix} + 2 \cdot \begin{bmatrix} -1 & 3 \\ -4 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -11 & -3 \\ 1 & -1 \end{bmatrix}$

F) Not defined, because A is a 3×2 matrix and C is a 2×2 matrix, so they do not have the same dimensions.

G) Not defined, because A is a 3×2 matrix and D is a 2×3 matrix, so they do not have the same dimensions.

H) $A - A = 0_{3\times 2}$

I) $C - C = 0_{2\times 2}$

J) $A, B \in \mathcal{M}_{3\times 2}(\mathbb{R})$. Since $2 \neq 3$, AB is undefined.

K) Given $A \in \mathcal{M}_{3\times 2}(\mathbb{R})$ and $C \in \mathcal{M}_{2\times 2}(\mathbb{R})$,

$$(AC)_{ij} = \sum_{k=1}^{2} a_{ik}c_{kj}, \quad i \in \{1,2,3\},\ j \in \{1,2\}.$$

$$(AC)_{11} = \sum_{k=1}^{2} a_{1k}c_{k1} = 2(2) + 2(3) = 10,$$

$$(AC)_{12} = \sum_{k=1}^{2} a_{1k}c_{k2} = 2(-1) + 2(0) = -2,$$

$$(AC)_{21} = \sum_{k=1}^{2} a_{2k}c_{k1} = 3(2) + 3(3) = 15,$$

$$(AC)_{22} = \sum_{k=1}^{2} a_{2k}c_{k2} = 3(-1) + 3(0) = -3,$$

$$(AC)_{31} = \sum_{k=1}^{2} a_{3k}c_{k1} = (-1)(2) + (-1)(3) = -5,$$

$$(AC)_{32} = \sum_{k=1}^{2} a_{3k}c_{k2} = (-1)(-1) + (-1)(0) = 1 \implies$$

$$\implies AC = \begin{bmatrix} 10 & -2 \\ 15 & -3 \\ -5 & 1 \end{bmatrix}$$

L) Given $C \in \mathcal{M}_{2\times 2}(\mathbb{R})$ and $A \in \mathcal{M}_{3\times 2}(\mathbb{R})$, the matrix product CA is undefined, because the number of columns of C is not equal to the number of rows of A.

M) Given $A \in \mathcal{M}_{3\times 2}(\mathbb{R})$ and $D \in \mathcal{M}_{2\times 3}(\mathbb{R})$,

$$(AD)_{ij} = \sum_{k=1}^{2} a_{ik} d_{kj}, \quad i \in \{1,2,3\},\ j \in \{1,2,3\}.$$

$$(AD)_{11} = \sum_{k=1}^{2} a_{1k} d_{k1} = 2(1) + 2(3) = 8,$$

$$(AD)_{12} = \sum_{k=1}^{2} a_{1k} d_{k2} = 2(2) + 2(1) = 6,$$

$$(AD)_{13} = \sum_{k=1}^{2} a_{1k} d_{k3} = 2(3) + 2(-2) = 2,$$

$$(AD)_{21} = \sum_{k=1}^{2} a_{2k} d_{k1} = 3(1) + 3(3) = 12,$$

$$(AD)_{22} = \sum_{k=1}^{2} a_{2k} d_{k2} = 3(2) + 3(1) = 9,$$

$$(AD)_{23} = \sum_{k=1}^{2} a_{2k} d_{k3} = 3(3) + 3(-2) = 3,$$

$$(AD)_{31} = \sum_{k=1}^{2} a_{3k} d_{k1} = (-1)(1) + (-1)(3) = -4,$$

$$(AD)_{32} = \sum_{k=1}^{2} a_{3k} d_{k2} = (-1)(2) + (-1)(1) = -3,$$

$$(AD)_{33} = \sum_{k=1}^{2} a_{3k} d_{k3} = (-1)(3) + (-1)(-2) = -1 \implies$$

$$\implies AD = \begin{bmatrix} 8 & 6 & 2 \\ 12 & 9 & 3 \\ -4 & -3 & -1 \end{bmatrix}$$

N) Given $D \in \mathcal{M}_{2\times 3}(\mathbb{R})$ and $A \in \mathcal{M}_{3\times 2}(\mathbb{R})$,

$$(DA)_{ij} = \sum_{k=1}^{3} d_{ik} a_{kj}, \quad i \in \{1,2\},\ j \in \{1,2\}.$$

$$(DA)_{11} = \sum_{k=1}^{3} d_{1k}a_{k1} = 1(2) + 2(3) + 3(-1) = 5,$$

$$(DA)_{12} = \sum_{k=1}^{3} d_{1k}a_{k2} = 1(2) + 2(3) + 3(-1) = 5,$$

$$(DA)_{21} = \sum_{k=1}^{3} d_{2k}a_{k1} = 3(2) + 1(3) + (-2)(-1) = 11,$$

$$(DA)_{22} = \sum_{k=1}^{3} d_{2k}a_{k2} = 3(2) + 1(3) + (-2)(-1) = 11 \implies$$

$$\implies DA = \begin{bmatrix} 5 & 5 \\ 11 & 11 \end{bmatrix}$$

O) Given $E \in \mathcal{M}_{3\times 3}(\mathbb{R})$ and $F \in \mathcal{M}_{3\times 3}(\mathbb{R})$,

$$(EF)_{ij} = \sum_{k=1}^{3} e_{ik}f_{kj}, \quad i \in \{1,2,3\},\ j \in \{1,2,3\} \implies$$

$$EF = \begin{bmatrix} -2 & 3 & 9 \\ 0 & 2 & -4 \\ 2 & 4 & 2 \end{bmatrix}$$

P) Given $F \in \mathcal{M}_{3\times 3}(\mathbb{R})$ and $E \in \mathcal{M}_{3\times 3}(\mathbb{R})$,

$$(FE)_{ij} = \sum_{k=1}^{3} f_{ik}e_{kj}, \quad i \in \{1,2,3\},\ j \in \{1,2,3\}.$$

$$FE = \begin{bmatrix} -9 & 8 & 9 \\ 2 & -2 & -9 \\ -7 & 8 & 13 \end{bmatrix}$$

Q) $A(3D)$. We know from the previous result on M that AD is defined, so $A(3D)$ is also defined. The unique change is that D is now multiplied by 3, and given the property of matrix multiplication:

$$\alpha(XY) = (\alpha X)Y = X(\alpha Y), \quad \alpha \in \mathbb{R},\ X \in \mathcal{M}_{m\times n}(\mathbb{R}),\ Y \in \mathcal{M}_{n\times p}(\mathbb{R}).$$

$\therefore$, we can write:

$$A(3D) = 3(AD) = 3 \cdot \begin{bmatrix} 8 & 6 & 2 \\ 12 & 9 & 3 \\ -4 & -3 & -1 \end{bmatrix} = \begin{bmatrix} 24 & 18 & 6 \\ 36 & 27 & 9 \\ -12 & -9 & -3 \end{bmatrix}$$

R) $(AC)D$. We know from the previous result on K that AC is defined. $AC \in \mathcal{M}_{3\times 2}(\mathbb{R})$ and $D \in \mathcal{M}_{2\times 3}(\mathbb{R}) \implies$ that $(AC) \cdot D$ is defined.

$$(AC)D = \begin{bmatrix} 10 & -2 \\ 15 & -3 \\ -5 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 4 & 18 & 34 \\ 6 & 27 & 51 \\ -2 & -9 & -17 \end{bmatrix}$$

S) CD is defined because C is 2×2 and D is 2×3; hence CD is 2×3 and $A(CD)$ is defined. By associativity of matrix multiplication and the result in R),

$$A(CD) = (AC)D = \begin{bmatrix} 4 & 18 & 34 \\ 6 & 27 & 51 \\ -2 & -9 & -17 \end{bmatrix}.$$

T) Let's call T to the matrix result of the evaluation of the expression in the line below.
By associativity of matrix multiplication and compatibility with scalar multiplication, we can regroup the factors and factor out the scalars:

$$T = (2A)(C(3D)) = 2 \cdot 3\big(A(CD)\big) = 6(AC)D$$

Since we know that $T = 6 \cdot (AC)D$, then:

$$\implies 6 \cdot \begin{bmatrix} 4 & 18 & 34 \\ 6 & 27 & 51 \\ -2 & -9 & -17 \end{bmatrix} = \begin{bmatrix} 24 & 108 & 204 \\ 36 & 162 & 306 \\ -12 & -54 & -102 \end{bmatrix}$$

U) By associativity and compatibility with scalar multiplication,

$$(2A)((2C)(2D)) = 8(AC)D = \begin{bmatrix} 32 & 144 & 272 \\ 48 & 216 & 408 \\ -16 & -72 & -136 \end{bmatrix}.$$

V) By distributivity and associativity, using R) and M),

$$A(CD + D) = (AC)D + AD = \begin{bmatrix} 12 & 24 & 36 \\ 18 & 36 & 54 \\ -6 & -12 & -18 \end{bmatrix}.$$

W)

$$C^2 = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 6 & -3 \end{bmatrix}.$$

Therefore,

$$C^3 = C^2C = \begin{bmatrix} 1 & -2 \\ 6 & -3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ 3 & -6 \end{bmatrix}.$$

5. Consider the following matrices:

$$A = \begin{bmatrix} 0 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 0 & 2 \end{bmatrix} \quad \textit{and} \quad B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 3 \\ 2 & 8 & 1 \end{bmatrix}$$

Without computing the product, calculate:

a) $(AB)_{23} = (-1)\cdot 0 + 0\cdot 3 + 2\cdot 1 = 2$

$(AB)_{31} = 3(-1) + 0\cdot 0 + 2\cdot 2 = 1.$

b) Second row of AB The second row of AB is the second row of A multiplied by B:

$$\begin{bmatrix}-1 & 0 & 2\end{bmatrix}\cdot B = \begin{bmatrix}5 & 16 & 2\end{bmatrix}$$

c) Third column of AB The third column of AB is A multiplied by the third column of B:

$$A\cdot\begin{bmatrix}0\\3\\1\end{bmatrix} = \begin{bmatrix}10\\2\\2\end{bmatrix}$$

6. Let the following matrices be defined:

$$A = \begin{bmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\\ d_1 & d_2 & d_3\end{bmatrix},\quad B = \begin{bmatrix}1 & 0 & 1 & 0\end{bmatrix},\quad C = \begin{bmatrix}1 & 0 & 1 & 0\\ 1 & -2 & 0 & 0\end{bmatrix},\quad D = \begin{bmatrix}1\\0\\1\end{bmatrix},\quad E = \begin{bmatrix}1 & 1 & 1\\ 0 & -2 & 0\\ 1 & 0 & 1\end{bmatrix}$$

Compute:

a) BA b) CA c) AD d) AE

Solution:

f) $B_{1\times 4}\cdot A_{4\times 3} \implies BA_{1\times 3} = \begin{bmatrix}a_1+c_1 & a_2+c_2 & a_3+c_3\end{bmatrix}$

f) $C_{2\times 4}\cdot A_{4\times 3} \implies CA_{2\times 3} = \begin{bmatrix}a_1+c_1 & a_2+c_2 & a_3+c_3\\ a_1-2b_1 & a_2-2b_2 & a_3-2b_3\end{bmatrix}$

f) $A_{4\times 3}\cdot D_{3\times 1} \implies AD_{4\times 1} = \begin{bmatrix}a_1+a_3\\ b_1+b_3\\ c_1+c_3\\ d_1+d_3\end{bmatrix}$

f) $A_{4\times 3}\cdot E_{3\times 3} \implies AE_{4\times 3} = \begin{bmatrix}a_1+a_3 & a_1-2a_2 & a_1+a_3\\ b_1+b_3 & b_1-2b_2 & b_1+b_3\\ c_1+c_3 & c_1-2c_2 & c_1+c_3\\ d_1+d_3 & d_1-2d_2 & d_1+d_3\end{bmatrix}$

7. Let the following matrices be defined:

$$A = \begin{bmatrix} 3 \\ 2 \\ -1 \\ 1 \end{bmatrix}, \qquad B = \begin{bmatrix} -1 & 0 & 1 & 2 \end{bmatrix}.$$

Compute AB and BA.

Solution:

Given $A \in \mathcal{M}_{4\times 1}(\mathbb{R})$ and $B \in \mathcal{M}_{1\times 4}(\mathbb{R})$, both products are defined.

$$AB = \begin{bmatrix} 3(-1) & 3(0) & 3(1) & 3(2) \\ 2(-1) & 2(0) & 2(1) & 2(2) \\ (-1)(-1) & (-1)(0) & (-1)(1) & (-1)(2) \\ 1(-1) & 1(0) & 1(1) & 1(2) \end{bmatrix} = \begin{bmatrix} -3 & 0 & 3 & 6 \\ -2 & 0 & 2 & 4 \\ 1 & 0 & -1 & -2 \\ -1 & 0 & 1 & 2 \end{bmatrix}.$$

$$\begin{aligned} BA &= \left[\sum_{k=1}^{4} b_{1k}a_{k1}\right] \\ &= \begin{bmatrix}(-1)(3) + 0(2) + 1(-1) + 2(1)\end{bmatrix} \\ &= \boxed{\begin{bmatrix}-2\end{bmatrix}}. \end{aligned}$$

8. Consider the matrices A and B from Exercise 5:

$$A = \begin{bmatrix} 0 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 0 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 3 \\ 2 & 8 & 1 \end{bmatrix}.$$

Determine all matrices X satisfying

$$2B^2 + \frac{1}{2}X - BA = (A - B)^2 + AB.$$

Solution:

Given $A, B \in \mathcal{M}_{3\times 3}(\mathbb{R})$, we seek $X \in \mathcal{M}_{3\times 3}(\mathbb{R})$.

Expanding $(A - B)^2$ without assuming $AB = BA$:

$$\begin{aligned} 2B^2 + \frac{1}{2}X - BA &= (A - B)^2 + AB \\ &= (A - B)(A - B) + AB \\ &= A^2 - AB - BA + B^2 + AB \\ &= A^2 - BA + B^2. \end{aligned}$$

Adding BA to both sides and isolating X:

$$
\begin{aligned}
2B^2 + \frac{1}{2}X &= A^2 + B^2 \\
\frac{1}{2}X &= A^2 + B^2 - 2B^2 \\
&= A^2 - B^2 \\
X &= 2(A^2 - B^2).
\end{aligned}
$$

Computing A^2 and B^2:

$$
\begin{aligned}
A^2 &= AA \\
&= \begin{bmatrix} 0 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 0 & 2 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 8 \\ 6 & -3 & 3 \\ 6 & 9 & 7 \end{bmatrix}.
\end{aligned}
$$

$$
\begin{aligned}
B^2 &= BB \\
&= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 3 \\ 2 & 8 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 3 \\ 2 & 8 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 6 & 25 & 0 \\ 0 & 0 & 25 \end{bmatrix}.
\end{aligned}
$$

Substituting:

$$
\begin{aligned}
X &= 2(A^2 - B^2) \\
&= 2\left(\begin{bmatrix} 0 & 0 & 8 \\ 6 & -3 & 3 \\ 6 & 9 & 7 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 6 & 25 & 0 \\ 0 & 0 & 25 \end{bmatrix} \right) \\
&= 2 \begin{bmatrix} -1 & 0 & 8 \\ 0 & -28 & 3 \\ 6 & 9 & -18 \end{bmatrix} \\
&= \boxed{\begin{bmatrix} -2 & 0 & 16 \\ 0 & -56 & 6 \\ 12 & 18 & -36 \end{bmatrix}}.
\end{aligned}
$$

Since the coefficient of X is nonzero, this solution is unique.

9. Let the following matrices be defined:

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}.$$

Compute AB, BA, and CA.
Solution:

Note: $A, B, C \in \mathcal{M}_{2\times 2}(\mathbb{R})$, so all three products are defined and belong to $\mathcal{M}_{2\times 2}(\mathbb{R})$.

Product AB:

$$AB = \begin{bmatrix} 4(-1)+2(2) & 4(-1)+2(2) \\ 2(-1)+1(2) & 2(-1)+1(2) \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}.$$

Product BA:

$$BA = \begin{bmatrix} (-1)4+(-1)2 & (-1)2+(-1)1 \\ 2(4)+2(2) & 2(2)+2(1) \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} -6 & -3 \\ 12 & 6 \end{bmatrix}}.$$

Product CA:

$$CA = \begin{bmatrix} 0(4)+(-3)2 & 0(2)+(-3)1 \\ 3(4)+0(2) & 3(2)+0(1) \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} -6 & -3 \\ 12 & 6 \end{bmatrix}}.$$

10. Let

$$A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Compute A^2, A^3, A^4, and A^5.

Solution:

$$A^2 = A \cdot A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Since $A^2 = A$, it's clearer that, by the definition of the matrix power:

$$\forall p \in \mathbb{N}^+, \quad A^p = A^{p-1} \cdot A \implies$$

Therefore:

$$\implies A^3 = \underbrace{A^2}_{A} \cdot A = A \cdot A = A^2 = A$$
$$\implies A^4 = \underbrace{A^3}_{A} \cdot A = A \cdot A = A^2 = A$$
$$\implies A^5 = \underbrace{A^4}_{A} \cdot A = A \cdot A = A^2 = A$$

Alternative solution: (extra)

Since

$$\boxed{A^2 = A}$$

we may avoid the efford of calculte the other powers using induction.

This means:

$$\boxed{\forall n \in \mathbb{N}^+, \quad A^n = A}$$

Base case ($n = 1$):

$$A^1 = A.$$

Inductive hypothesis:

Let $n \in \mathbb{N}_{>0}$ and assume that $A^n = A$.

Inductive step:

Multiplying both sides of the inductive hypothesis by A on the right:

$$A^n = A \implies A^n A = AA.$$

Therefore,

$$\begin{aligned} A^{n+1} &= A^n A \\ &= AA \\ &= A^2 \\ &= A. \end{aligned}$$

Therefore, $P(n) \implies P(n+1) \implies \boxed{A^n = A}$

11. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}.$$

Verify that there exists a power of A which is the zero matrix.

Solution:

Given, $A \in \mathcal{M}_3(\mathbb{R})$ they are asking for:

$$\exists k \in \mathbb{N}+ : A^k = 0_3.$$

Computing A^2:

$$\begin{aligned} A^2 &= AA \\ &= \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \neq 0_3. \end{aligned}$$

Computing A^3:

$$\begin{aligned} A^3 &= A^2 A \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}. \end{aligned}$$

Therefore,

$$\boxed{\exists k = 3 \in \mathbb{N}^+ : A^k = 0_3}.$$

12. Consider the matrices

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Compute $(AB)^2$, A^2B^2, and B^2A^2.

Solution:

Given $A, B \in \mathcal{M}_{2\times 2}(\mathbb{R})$, all three products are defined.

Product $(AB)^2$:

First, compute AB:

$$AB = \begin{bmatrix} 1(1)+1(-1) & 1(1)+1(1) \\ 1(1)+0(-1) & 1(1)+0(1) \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}$$

Therefore,

$$(AB)^2 = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}}$$

Product A^2B^2:

Computing both squares:

$$A^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad B^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

Then,

$$A^2B^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = \boxed{\begin{bmatrix} -2 & 4 \\ -2 & 2 \end{bmatrix}}$$

Product B^2A^2:

$$B^2A^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 2 & 2 \\ -4 & -2 \end{bmatrix}}$$

Note:

$$\boxed{(AB)^2 \neq A^2B^2 \neq B^2A^2}$$

In general, $(AB)^2 = ABAB$, but $ABAB \neq A^2B^2$ (without assuming $AB = BA$).

13. **State, with justification, whether the following statements are true or false.**

Let $n \in \mathbb{N}^+$ and $A, B, C \in \mathcal{M}_n(\mathbb{K})$.

a) **For any square matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of the same order, $\boldsymbol{AB = BA}$.**
Universal statement:

$$\forall A, B \in \mathcal{M}_n(\mathbb{K}) : AB = BA.$$

False. Proof by counterexample.
Consider

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then,

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = BA.$$

Therefore, the statement is false.

Sufficient condition:
If A and B are diagonal matrices, then, for all $i, j \in \{1, \ldots, n\}$,

$$(AB)_{ii} = a_{ii}b_{ii} = b_{ii}a_{ii} = (BA)_{ii},$$
$$(AB)_{ij} = 0 = (BA)_{ij}, \quad i \neq j.$$

Hence, all corresponding entries are equal, so

$$\boxed{AB = BA}.$$

b) **For any square matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of the same order, if $\boldsymbol{AB = 0}$, then $\boldsymbol{A = 0}$ or $\boldsymbol{B = 0}$.**
Universal statement:

$$\forall A, B \in \mathcal{M}_n(\mathbb{K}) : AB = 0_n \implies (A = 0_n \vee B = 0_n).$$

False. Proof by counterexample.
Consider

$$A = B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0_2.$$

However,

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0_2$$

Thus,

$$AB = 0_2 \quad \wedge \quad A \neq 0_2 \quad \wedge \quad B \neq 0_2.$$

Therefore, the statement is false.

Sufficient condition:

If either A or B is invertible, the implication holds.

Case 1: A is invertible.

Multiplying by A^{-1} on the left:

$$\begin{aligned} AB = 0_n &\implies A^{-1}(AB) = A^{-1}0_n \\ &\implies (A^{-1}A)B = 0_n \\ &\implies I_nB = 0_n \\ &\implies \boxed{B = 0_n}. \end{aligned}$$

Case 2: B is invertible.

Multiplying by B^{-1} on the right:

$$\begin{aligned} AB = 0_n &\implies (AB)B^{-1} = 0_nB^{-1} \\ &\implies A(BB^{-1}) = 0_n \\ &\implies AI_n = 0_n \\ &\implies \boxed{A = 0_n}. \end{aligned}$$

Hence, if at least one factor is invertible, the other must be zero whenever $AB = 0_n$.

c) **For any square matrices A, B, and C of the same order, if $AB = AC$, then $B = C$.**

Universal statement:

$$\forall A, B, C \in \mathcal{M}_n(\mathbb{K}) : AB = AC \implies B = C.$$

False. Proof by counterexample.

Consider

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Then,

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = AC,$$

while $B \neq C$.

Therefore, the statement is false.

Sufficient condition:

If A is invertible, there exists $A^{-1} \in \mathcal{M}_n(\mathbb{K})$ such that

$$A^{-1}A = AA^{-1} = I_n.$$

Multiplying both sides of $AB = AC$ by A^{-1} on the left:

$$\begin{aligned} AB = AC &\implies A^{-1}(AB) = A^{-1}(AC) \\ &\implies (A^{-1}A)B = (A^{-1}A)C \\ &\implies I_n B = I_n C \\ &\implies \boxed{B = C}. \end{aligned}$$

Hence,

$$\boxed{A \text{ invertible} \wedge AB = AC \implies B = C.}$$

d) **If, for a square matrix A, there exists $k \in \mathbb{N}$ such that $A^k = 0$, then $A = 0$.**

Universal statement:

$$\forall A \in \mathcal{M}_n(\mathbb{K}) : \left(\exists k \in \mathbb{N}^+ : A^k = 0_n\right) \implies A = 0_n.$$

False. Proof by counterexample.

Consider

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0_2.$$

For $k = 2$,

$$A^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = 0_2$$

Thus,

$$\exists k = 2 \in \mathbb{N}^+ : A^k = 0_2, \quad A \neq 0_2.$$

Therefore, the statement is false.

Sufficient condition:

Suppose A is idempotent, i.e.,

$$A^2 = A.$$

By mathematical induction,

$$\forall p \in \mathbb{N}^+ : A^p = A.$$

Indeed, the base case $A^1 = A$ holds. Assuming $A^p = A$ for some $p \geq 1$, multiplying by A gives

$$\begin{aligned} A^{p+1} &= A^p A \\ &= AA \\ &= A^2 \\ &= A. \end{aligned}$$

Now, if

$$\exists k \in \mathbb{N}^+ : A^k = 0_n,$$

then, since $A^k = A$,

$$A = A^k = 0_n.$$

Therefore,

$$\boxed{A = 0_n}.$$

A matrix that is both idempotent and nilpotent must be zero.

e) **For any square matrices A and B of the same order, $(AB)^2 = A^2B^2$.**
Universal statement:

$$\forall A, B \in \mathcal{M}_n(\mathbb{K}) : (AB)^2 = A^2B^2.$$

False. Proof by counterexample.
Consider

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Computing AB,

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Therefore,

$$(AB)^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

However,

$$A^2 = 0_2, \qquad B^2 = 0_2,$$

so

$$A^2B^2 = 0_2.$$

Hence,

$$(AB)^2 \neq A^2B^2,$$

and the statement is false.

Sufficient condition:
Suppose $AB = BA$.
Then,

$$\begin{aligned}(AB)^2 &= (AB)(AB)\\ &= A(BA)B\\ &= A(AB)B\\ &= (AA)(BB)\\ &= \boxed{A^2B^2}.\end{aligned}$$

Therefore,

$$\boxed{AB = BA \implies (AB)^2 = A^2B^2.}$$

14. **Let $A_1, A_2, \ldots, A_n$ be matrices such that the product $A_1A_2\cdots A_n$ is defined.**

Express $(A_1A_2\cdots A_n)^T$ in terms of $A_1^T, A_2^T, \ldots, A_n^T$.

Using the transpose property

$$(AB)^T = B^TA^T,$$

and applying it successively, we obtain:

$$\begin{aligned}(A_1A_2\cdots A_n)^T &= ((A_1A_2\cdots A_{n-1})A_n)^T\\ &= A_n^T(A_1A_2\cdots A_{n-1})^T\\ &= A_n^TA_{n-1}^T(A_1A_2\cdots A_{n-2})^T\\ &\vdots\\ &= \boxed{A_n^TA_{n-1}^T\cdots A_2^TA_1^T}.\end{aligned}$$

So the transpose of a matrix product is the product of the transposed matrices in reverse order.

15. **Show that, for any square matrix A, the matrix $A + A^T$ is symmetric.**

By definition, a matrix A is symmetric if and only if $A = A^T$.

Let $A \in \mathcal{M}_n(\mathbb{K})$.
Note that $A^T \in \mathcal{M}_n(\mathbb{K})$, so $A + A^T$ is defined.

Using the properties of transposition:

$$\begin{aligned}(A + A^T)^T &= A^T + (A^T)^T\\ &= A^T + A\\ &= A + A^T \implies\\ \implies &\boxed{(A + A^T)^T = A + A^T}.\end{aligned}$$

16. Final item. Remove when done.